A Bialgebra Framework for Positivity in Schubert Calculus

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Plan

- Slightly expand the polynomial ring
- Dual algebra and basis indexing.
- Schubert, key, and forest bases in one transition framework.
- Final point: positivity questions through structure constants.

Direct sum of polynomial algebras

Define

$$\mathbf{S} = \bigoplus_{n \geq 0} \mathbf{S}_n, \qquad \mathbf{S}_n = \mathbb{Q}[x_1, \ldots, x_n].$$

Multiplication in $\mathbf{S}$ is componentwise:

- if $f, g \in \mathbf{S}_n$, then $f \cdot g$ is the usual product in $\mathbb{Q}[x_1, \ldots, x_n]$;
- if $f \in \mathbf{S}_m$, $g \in \mathbf{S}_n$ with $m \neq n$ and $m, n > 0$, then $f \cdot g = 0$.

So each graded piece is a familiar polynomial ring, and cross-component products are trivial. The coproduct is

$$\Delta(x_c) = \sum_{ab=c} x_a \otimes x_b$$

where ab is the concatenation of weak compositions. This makes $\mathbf{S}$ into a graded bialgebra.

$$\Delta(x_1^2 x_3 x_5) = 1 \otimes x_1^2 x_3 x_5 + x_1^2 \otimes x_3 x_5 + x_1^2 x_3 \otimes x_5 + x_1^2 x_3 x_5 \otimes 1.$$

Double grading

In addition to the degree as a polynomial, $\mathbf{S}$ is also graded by inclusion in the direct summand $\mathbf{S}_n$, which we as interpret as length.
We define the dual algebra $\mathcal{DS}$ to be the graded dual of $\mathbf{S}$ with respect to this $\mathbb{N} \times \mathbb{N}$ grading. For the monomial basis, which is indexed by weak compositions, we define the dual basis $[\alpha]$ for weak compositions α by

$$\langle x_\alpha, [\beta] \rangle = \delta_{\alpha,\beta}$$

Hence $\mathcal{DS}$ is also graded by length and degree, with the latter being the sum of the composition, and the product and coproduct are dual to those of $\mathbf{S}$ with respect to the degree grading.

Dual indexing and product

We have that the dual product is concatenation:

$$[\alpha] \cdot [\beta] = [\alpha\, \beta]\,.$$

Hence $\mathcal{DS}$ is a free associative algebra (tensor algebra on the generators $[k]$, $k \geq 0$). The coproduct is

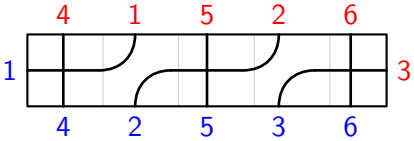
$$\Delta([\gamma]) = \sum_{\alpha+\beta=\gamma} [\alpha] \otimes [\beta]\,.$$

We now define and investigate a curious combinatorially defined basis of $\mathcal{DS}$ indexed by code-length bounded permutations.

Crossing diagrams

Think of a one-row strip of $n-1$ boxes.

- Label the left edge by 1.
- Label the bottom by a permutation of the numbers $2, 3, \ldots, n$ (left to right).
- In each box, place either a *cross* ⊞ or a *bump* ⧄.



Follow the strands through the row. The admissibility condition is that it cannot undo inversions, only create new ones.

A nice combinatorial basis

Let Ξ_w^k be the basis indexed by permutations w such that $w(p) < w(p+1)$ for all $p > k$ satisfying the following:

- $\Xi_w^1 = [w(1) - 1]$
- For a nonnegative integer $[p]$, we have that

$$[p] \cdot \Xi_w^k = \sum_{w'} \Xi_{w'}^{k+1}$$

  where w' is obtained from w through a valid crossing diagram with p crosses after adding 1 to each element of w and adding a new first element 1.

I can provide an explicit (but mildly cancellative) formula for Ξ_w^k in the word basis, as well as a positive formula for the product of Ξ_u^p and Ξ_v^q in terms of crossing diagrams. The coproduct structure constants are also nonnegative integers, but I do not have a positive combinatorial formula for them yet...

Taking off the mask: The dual Schubert basis

The basis satisfies

$$\langle \mathfrak{S}_u(x_1, \ldots, x_n), \Xi_v^n \rangle = \delta_{u,v}$$

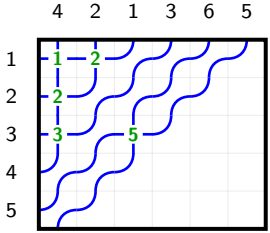
where $\mathfrak{S}_u(x_1, \ldots, x_n)$ is the Schubert polynomial corresponding to the permutation u.
The coefficients $c_{u,v}^w$ of the coproduct formula

$$\Delta\, \Xi_w^n = \sum_{u,v} c_{u,v}^w\, \Xi_u^n \otimes \Xi_v^n$$

are the Littlewood-Richardson coefficients for the complete flag variety on a complex vector space of dimension $n+1$.
This is the main structural link: Schubert positivity data appears as bialgebra structure constants.
(Yes, the crossing diagrams are secretly pipe dreams.)

Bounded pipe dreams

Pipe dreams are well-studied objects introduced in the 1990s by Bergeron and Billey to give a nicer combinatorial formula for Schubert polynomials equivalent to an earlier one by Billey, Jockusch, and Stanley involving reduced words/compatible sequences. Bounded pipe dreams have an additional length parameter tacked on.



Theorem (Bergeron, Billey, 1993)

The Schubert polynomial $\mathfrak{S}_w$ is equal to

$$\sum_{R \in \mathrm{Pipe}(w)} x_{\mathrm{wt}(R)}$$

Transition algorithm on bounded pipe dreams

There is an algorithm that takes a bounded pipe dream of length n with row n empty to a unique pipe dream of length $n-1$ in a way that matches the Schubert transition formula in a strong, weight-preserving, bijective form.

Input: A bounded pipe dream R with $\mathrm{length}(R) = n$ and row n empty.
Output: A bounded pipe dream R' with $\mathrm{length}(R') = n-1$, $|R'| = |R|$ (same number of crossing in each row), and $w_R \overset{n}{\searrow} w_{R'}$.
if R with last row removed is a valid bounded pipe dream **then**
    return R with last row removed
else
    Locate crossings corresponding to maximal right descents of $w(R)$ until the maximal right descent is $n-1$, say at rows $i_1 < i_2 < \cdots < i_p$ with corresponding columns $j_1, j_2, \ldots, j_p$.
    $R^- \leftarrow R \setminus \{(i_1, j_1), \ldots, (i_p, j_p)\}$.
    Let $I \leftarrow (i_1, i_2, \ldots, i_p)$.
    $D \leftarrow R^- \cup \{(n,1), (n,2), \ldots, (n,p)\}$.
    $D' \leftarrow I \overset{n-1}{\rightsquigarrow} D$ (an insertion algorithm similar to existing ones for pipe dreams)
    $R' \leftarrow \{(i,j) \in D' \mid i < n\}$ as a bounded pipe dream with $n-1$ rows.
    return R'.
end if

Bounded pipe dream ring product

Given bounded pipe dreams R_1, R_2, define $R_1 \star R_2$ by taking the sum of all R obtained by

- *untransitioning* R_1 in all possible valid ways to create space while remaining a pipe dream,
- shifting R_2 upward,
- placing the shifted R_2 into the new empty region, unchanged, if doing so yields a valid pipe dream.

This allows us to lift the product of the dual algebra $\mathcal{DS}$ to a product on formal linear combinations of bounded pipe dreams.

Theorem (S, 2026+)

The product $\star$ is associative and gives a well-defined ring structure on the vector space Pipe *spanned by bounded pipe dreams.*

Pipe dream ring as a lift of $\mathcal{DS}$

There are two surjective linear maps from the pipe dream ring $\mathcal{R}$ to the dual algebra $\mathcal{DS}$:

$$\pi_{\mathrm{Sch}} : \mathcal{R} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{Sch}}(G) = \Xi^{(n)}_{\mathrm{perm}(G)},$$

$$\pi_{\mathrm{wt}} : \mathcal{R} \twoheadrightarrow \mathcal{DS}, \qquad \pi_{\mathrm{wt}}(G) = [\mathrm{wt}(G)]\,.$$

By virtue of the fact that $\mathcal{DS}$ is free, there is a section $\iota : \mathcal{DS} \hookrightarrow \mathcal{R}$ of π_{wt} that sends $[i]$ for an integer i to the unique pipe dream r_i with $\mathrm{wt}(r_i) = (i)$. It turns out that these are all ring homomorphisms and ι is also a section of π_{Sch}.

Theorem (S, 2026+)

We have that π_{Sch} and π_{wt} are surjective ring homomorphisms to $\mathcal{DS}$ and ι is an injective ring homomorphism from $\mathcal{DS}$ to $\mathcal{R}$ such that

$$\pi_{\mathrm{Sch}} \circ \iota = \pi_{\mathrm{wt}} \circ \iota = \mathrm{id}_{\mathcal{DS}}.$$

Immediate consequence

We immediately obtain a formula for the product of dual Schubert elements.

Theorem (S, 2026+)

For dual Schubert elements $\Xi_u^{n_1}$, $\Xi_v^{n_2}$, we have

$$\Xi_u^{n_1} \cdot \Xi_v^{n_2} = \sum_w d_{u,v}^w(n_1, n_2)\, \Xi_w^{n_1+n_2},$$

where, fixing pipe dreams R_u for u with $\mathrm{length}(R_u) = n_1$ and R_v for v with $\mathrm{length}(R_v) = n_2$, $d_{u,v}^w(n_1, n_2)$ is the number of bounded pipe dreams $R_w \in \mathrm{Pipe}(w)$ such that R_w has a coefficient of 1 in the product $R_u \star R_v$.
Equivalently,

$$\mathfrak{S}_w(x_1, \ldots, x_{n_1+n_2}) = \sum_{u,v} d_{u,v}^w(n_1, n_2)\, \mathfrak{S}_u(x_1, \ldots, x_{n_1})\mathfrak{S}_v(x_{n_1+1}, \ldots, x_{n_1+n_2}).$$

Little bumps

Suppose $(w_1, w_2, \ldots, w_n)$ is a reduced word for a permutation w. A *Little bump* is an operation that takes a reduced word for w together with an index $1 \leq i \leq n$ and produces a reduced word for a different permutation w' such that $\ell(w) = \ell(w')$. The original purpose of this definition by Little was to generalize the RSK correspondence to reduced words (in a manner distinct from Edelman-Greene insertion). However, the map has interesting combinatorial properties in and of iteslf.

Theorem (Hamaker, Young 2014)

Little bumps preserve the Coxeter-Knuth equivalence class of a reduced word.

Little bumps and the transition algorithm

The transition algorithm on pipe dreams is equivalent to a specific sequence of Little bumps on the corresponding reduced word that preserves compatibility with the weight. In particular, we have

Theorem (S, 2026+)

If R is a bounded pipe dream with $\mathrm{length}(R) = n$ and row n empty, then the transition algorithm is equivalent to applying Little bumps to the last descent of the permutation until the result has its last descent $\leq n - 1$.

Positivity theorem

Theorem (S, 2026+)

Suppose there is an equivalence relation on reduced words $\equiv$ such that the following hold:
If W, W' are reduced words such that $W \equiv W'$, then

- $\uparrow W \equiv \uparrow W'$, *where $\uparrow$ is the operation of adding 1 to each letter.*
- $\mathrm{perm}(W) = \mathrm{perm}(W')$.
- *If i is the last descent of $\mathrm{perm}(W)$, then $L_{i,i+1}(W) \equiv L_{i,i+1}(W')$.*

Then there is a unique family of polynomials with nonnegative integer coefficients that form a basis for **S** *that is the sum of monomials corresponding to pipe dreams with reduced words in the equivalence class. The dual basis is then positive in the dual Schubert basis and has nonnegative structure constants.*

Examples

This applies to the following examples of equivalence relations on reduced words:

- Schubert polynomials: $\mathrm{perm}(W) = \mathrm{perm}(W')$.
- Fundamental slide polynomials ($W \equiv W'$ iff $W = W'$)
- Key polynomials: $W \equiv W'$ if W and W' are Coxeter-Knuth equivalent.

Forest polynomials

Forest polynomials $\mathfrak{P}_F$ are a more recently discovered basis of $\mathbf{S}$ in which the Schubert basis also expands positively. They are indexed by objects F known as "indexed forests," which are canonically labeled forests of binary trees. Given a word W, there is an insertion algorithm that yields a labeled indexed forest denoted $\Omega(W)$. We then say that $\Omega(R)$ has shape F if the indexed forest obtained from R by this insertion algorithm is F.

Theorem (S, 2026+)

Let F be an indexed forest with $\mathfrak{c}(F) = \mathfrak{c}(w)$ for a permutation w. Then

$$\mathfrak{P}_F = \sum_{\substack{R \in \mathrm{Pipe}(w) \\ \Omega(R) \text{ has shape } F}} \mathrm{wt}(R)$$

Dual product of forest polynomials

Theorem (S, 2026+)

For each positive integer k, forest polynomials with $n \geq k$ variables satisfy the formula

$$\mathfrak{P}_F = \sum_{\gamma} d^F_{G,H}(k)\, \mathfrak{P}_G(x_1, \ldots, x_k) \mathfrak{P}_H(x_{k+1}, \ldots, x_n),$$

Where, fixing bounded pipe dreams R_G, R_H with $\mathrm{length}(R_G) = k$, $\mathrm{length}(R_H) = n - k$, $\mathrm{shape}(\Omega(R_G)) = G$ and $\mathrm{shape}(\Omega(R_H)) = H$, $d^H_{F,G}(k)$ is the number of bounded pipe dreams R with $\mathrm{length}(R) = n$ and $\mathrm{shape}(\Omega(R)) = F$ such that R has a coefficient of 1 in the product $R_G \star R_H$.